

COMPREHENSIVE_UNFINISHED_WORK_REP

Comprehensive Unfinished Work Report & Gap Analysis

Date: December 10, 2025 **Project:** HelixCode **Current Status:** 85-90% Complete
(Production Ready for Core Features)

1. Executive Summary

HelixCode is a robust, distributed AI development platform. While core functionalities (Server, Auth, Task Management, Worker Pools, Basic LLM Support) are production-ready, significant gaps exist in enterprise provider support, advanced tool capabilities, and comprehensive test coverage. This report details all unfinished work required to reach 100% completion.

2. Critical Feature Gaps

2.1. Missing LLM Providers (High Priority)

The following industry-standard providers are currently missing or incomplete, limiting HelixCode's competitiveness against agents like Claude Code or OpenCode. * **Anthropic (Claude): CRITICAL.** Missing direct API integration. Required for "Extended Thinking", Prompt Caching, and Claude 3.7 Sonnet/Opus support. * **Google Gemini: CRITICAL.** Missing direct integration. Required for 1M+ token context windows and native multimodal capabilities. * **AWS Bedrock:** Missing. Required for enterprise compliance and security. * **Azure OpenAI:** Missing. Required for Microsoft-centric enterprise environments. * **Vertex AI:** Missing. Google Cloud enterprise integration. * **Groq:** Missing. Required for ultra-low latency inference. * **Mistral:** Missing direct API support.

2.2. Incomplete Memory Providers

While Zep, ChromaDB, Pinecone, Qdrant, and FAISS are implemented, the following are stubs or missing: * **Weaviate:** Stubs exist but methods are unimplemented (15+ TODOs). * **Mem0:** Not implemented. * **Memonto:** Not implemented. * **BaseAI:** Not implemented.

2.3. Tool System Deficiencies

Compared to competitors, the tool system lacks: * **File System Tools:** Comprehensive read/write/search/diff tools are not fully exposed to agents. * **Shell Execution:** Safe, sandboxed shell execution is limited. * **Web Tools:** No integrated web search or URL fetching/parsing. * **Multi-File Editing:** Atomic multi-file editing transactions are missing.

2.4. UI & Configuration

- **Terminal UI:** Missing “New Task” form and Cognition integration controls.
- **Configuration API:** Missing endpoints for runtime config updates (RestoreConfig, ReloadConfig) and WebSocket handlers.
- **Model Management:** Missing tools for converting models (GGUF/GGML) and usage analytics dashboards.

3. Testing Gaps & Technical Debt

3.1. Low Coverage Modules

- **internal/task: 28.6% Coverage.** Critical module. Needs comprehensive database mock tests.
- **internal/auth: 47.0% Coverage.** Needs more integration tests and edge case handling.
- **internal/deployment: 15.0% Coverage.**
- **internal/cognition: 12.5% Coverage.**

3.2. Missing Test Types

- **Integration Tests:** Missing for new/planned providers (Anthropic, Gemini).
- **E2E Tests:** Full workflow tests for enterprise providers are impossible until implemented.
- **Performance Tests:** Need benchmarks for new providers and high-load scenarios.

4. Documentation & Educational Content Gaps

- **Website:** The Website directory is missing or empty. Content needs to be created from scratch.
- **Video Courses:** No video content exists. Scripts and curriculum need to be developed.
- **User Manuals:** Existing manuals need updates to cover new providers and advanced features.

5. Summary of “TODO” Items

- ~70 **TODOs** scattered across the codebase (e.g., `internal/config/config_api.go`, `internal/llm/model_download_manager.go`).
 - **Legacy Code:** `external/memory/zep/legacy/` contains unresolved TODOs.
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Conclusion: To achieve 100% completion, the project requires a focused 10-week implementation plan addressing these specific gaps.